

SIMATS ENGINEERING
ASSESSMENT TOOL 1: Short Answer Test

COURSE NAME: Cloud Computing and
Big Data Analytics

COURSE CODE: CSA1516

COURSE OUTCOME COVERED

CO1: Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions. (BL2)

Assessment Tool Weightage: 40%

Dave's Taxonomy: Imitation (Level 1)

Duration: 60 Minutes

Total Marks: 25

Question Count: 5

Code of Conduct

I Y. Naga Gixidhar (C192572124) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Y.N.Gixidhar.

Criteria	Conceptual Accuracy (1.5)	Coverage of Concepts (1)	Use of Examples (0.75)	Comparison / Differentiation (1)	Clarity of Expression (0.75)	TOTAL (5)
Weightage	30%	20%	15%	20%	15%	100%
Q1						
Q2						
Q3						
Q4						
Q5						

Signature of the Faculty

Total Marks Awarded:

Assessment Tasks

Short Answer Test

1. Describe the evolution from standalone computing to cloud-based systems. Summarize key technological transitions that enabled cloud computing.
2. Interpret the working of hypervisors in virtualization. Differentiate between Type 1 and Type 2 hypervisors with examples.
3. Describe the role of APIs in cloud service integration. Explain how APIs support interoperability between services..
4. Explain the concept of resource pooling in cloud computing. Illustrate how it benefits multiple users.
5. Interpret the concept of rapid elasticity in cloud computing. Explain how it supports dynamic workloads.

Short Answer Test Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Conceptual Accuracy	30%	Accurate definitions and precise explanation of all key terms	Mostly accurate with minor gaps	Partial understanding with noticeable errors	Incorrect or irrelevant explanation
Coverage of Concepts	20%	Covers all expected sub-points completely	Covers most expected points	Covers only basic points	Very limited coverage
Use of Examples	15%	Uses highly relevant cloud examples to strengthen the answer	Uses relevant examples with minor mismatch	Uses vague or partially relevant examples	No useful examples
Comparison / Differentiation	20%	Clearly distinguishes concepts with strong logic	Reasonable differentiation with minor overlap	Comparison is weak or incomplete	Fails to differentiate concepts

Clarity of Expression	15%	Clear, organized, and concise answer writing	Generally clear with few issues	Understandable but poorly organized	Difficult to follow
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1) Evolution from standalone computing to cloud-based systems

The evolution of computing began with standalone computers, where applications & data were stored and processed on a single machine. Organizations later adopted client-server computing, allowing multiple users to access centralized servers over a network. This was followed by distributed computing.

The emergence of virtualization enabled multiple virtual machines to run on a single physical server. High speed internet, broadband connectivity, web services and large scale data centers further supported the development of cloud computing.

Key technological transitions :-

- * High-speed internet
- * Virtualization
- * Data centers
- * Web services
- * Broadband networking.

2A) A hypervisor is software that creates and manages multiple virtual machines (VMs) on a single physical computer.

Type 1 Hypervisor	Type 2 Hypervisor
→ Runs directly on hardware → Faster and more secure → used in data centers → EX:- VMware ESX, Microsoft Hyper-V	→ Runs on top of an operating system → Easier to install but slower. → used for personal computers. → EX:- Oracle VirtualBox, VMware Workstation.

3A) Role of APIs in cloud service integration :-

- * APIs (Application Programming Interfaces) allow different cloud applications and services to communicate.
- * They enable data sharing, automation and integration
- * APIs help different cloud platforms work together regardless of programming language or operating system
- * Example:- A mobile app using a cloud storage API to upload files.

Resource Pooling in cloud computing :-

- Resource Pooling is one of the essential characteristics of cloud computing. It refers to the practice of combining physical and virtual computing resources - such as servers, storage, memory, network bandwidth, and processing power - into a shared pool. These pooled resources are dynamically assigned and re-assigned to multiple users based on their requirements without the users knowing the exact physical location of the resources.

cloud providers use a multi-tenant architecture, where many customers securely share the same infrastructure while keeping their data and applications isolated.

Benefits of Resource Pooling :-

- * Handles sudden traffic spikes
- * Reduces costs by releasing unused resources
- * Improves application performance
- * supports dynamic and changing workload.

Example :- An online shopping website automatically adds more servers during a festival sale and removes them after the traffic decreases.

5A) Rapid Elasticity in cloud computing:-

↳ Rapid elasticity is a key feature of cloud computing that allows computing resources such as CPU, memory, storage, and network capacity to be increased or decreased automatically according to user demand.

Resources can be provisioned within minutes or even seconds, making cloud computing services highly flexible.

↳ Rapid elasticity helps organizations handle dynamic workloads, where demand changes frequently. During periods of high traffic, additional resources are automatically allocated to maintain performance.

Advantages:-

- * Ensures high performance during peak usage.
- * Prevents system overload and downtime.
- * Enables organizations to respond quickly to changing business needs.